SYSTEM MINDMAP & WORKFLOW ARCHITECTURE

Spare Part Control & Equipment Repair Management Web Application

1. APPLICATION MODULE MINDMAP

A. Inventory & Barcode/QR Control

* Master Spare Part List & Location Tracking
* Real-time Stock Balance & Min/Max Limit Alerts
* Barcode & QR Code Tag Generation & Scanner
* Part Unit Price & Valuation Calculation

B. Equipment & Asset Management

* Equipment Registry (Asset Code, Price, Budget)
* Main Machine vs Spare Machine Classification
* Batch Printing Equipment QR Code Stickers
* Individual Machine Service & Repair History

C. Transaction & Repair Withdrawal

* Quick Spare Part Withdrawal & Restock
* Link Withdrawal directly to Machine Repairs
* Auto Cost Allocation (Part Cost + Labor Cost)
* Instant Telegram Withdrawal Notifications

D. 12-Month PM Inspection & Checklist

* 4 Factory Forms (Strapping, Nail Gun, Seal, etc.)
* Inline Template Editor & Check Item Customization
* Friction Weld Seam Visual Standard Guidelines
* 3-Level Signatures (Tech, Engineer, Manager)

E. Tool Loan & Spare Swap Management

* Line Machine Spare Swapping & Borrowing Log
* Borrower, Location, Expected Return Date
* Return Tracking & Condition Inspection Notes
* Instant Telegram Borrow/Return Notifications

F. Google Sheets & Cloud Persistence

* 2-Way Real-time Google Sheets Cloud Sync
* AuthorizedUsers Role Access Control (Admin/User)
* Automatic PMInspections & Logs Provisioning
* Local Storage + Cloud Fallback Architecture

2. CORE SYSTEM WORKFLOW & DATA FLOW

STEP 1: User Login & Google Sheets Database Sync

User authenticates via Google OAuth -> Verifies AuthorizedUsers email -> Auto-fetches/provisions Sheet tabs.

STEP 2: Machine Breakdown / Spare Part Withdrawal Request

Tech scans Machine QR -> Selects required Spare Part -> Records Withdrawal & Repair Detail -> Auto Stock Deduction.

STEP 3: Monthly Preventive Maintenance (PM) Inspection

Selects Machine Form Template -> Conducts 12-Month Matrix Check -> Auto-Saves -> 3-Level Digital Verification.

STEP 4: Real-time Notifications & Analytics Sync

Triggers Telegram alerts (Withdrawal / Low Stock / Borrow) -> Updates Executive Dashboard Analytics in real-time.